

1 Systems of linear eqs

Definition (Consistent vs. Inconsistent). A system of linear eqs can have three things happen regarding its solutions. Specifically, the system can have:

- (1) No solutions;
- (2) A single/unique solution;
- (3) Infinitely many solutions.

If the system has *no solutions*, we say the system is inconsistent. If the system has either a *unique solution* or *infinitely many solutions*, then we call the system consistent.

Definition. In any matrix, the first *nonzero* entry of a row is called a pivot. If a row has a pivot then it is a “pivot row”, and if a column has a pivot it is a “pivot column”. Thus, a row of all 0’s isn’t a pivot row.

Definition ((Reduced) Echelon Form). There are four basic conditions:

- (i) Entries *beneath* pivots are all 0;
- (ii) Rows of 0’s (if there are any) moved to bottom;
- (iii) Entries *above* pivots are all 0;
- (iv) All pivot entries are equal to 1.

- If (at least) (i) and (ii) hold, then matrix is in echelon form (EF).
- If (i), (ii), (iii), and (iv) *all* hold, then matrix is in reduced echelon form (REF).

Note: Any matrix “has” (or “can-be-reduced-to”) an (EF), not just augmented matrices. A matrix can have many different (EF)’s, but it will only have one (REF).

Definition (Row Operations). In reducing matrices to (EF) or (REF) there are three *row operations* we can use:

- (1) Swap two rows: $(R_i \leftrightarrow R_j)$

$$(R_1 \leftrightarrow R_3) : \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \longrightarrow \begin{bmatrix} 7 & 8 & 9 \\ 4 & 5 & 6 \\ 1 & 2 & 3 \end{bmatrix};$$

- (2) Scale a row: $(R_i \rightarrow cR_i)$

$$(R_3 \rightarrow \tfrac{1}{2}R_3) : \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 14 & 16 & 18 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix};$$

- (3) Add a (scaled-)copy of one row to another: $(R_i \rightarrow R_i + cR_j)$

$$(R_3 \rightarrow R_3 - 7R_1) : \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 0 & -6 & -12 \end{bmatrix}.$$

Note: The matrices I put above are all just “plain” matrices—they are not “augmented” matrices b/c we aren’t solving a system of linear eqs here.

How to solve a system of linear eqs.

To solve a system of linear eqs, we *always* use the first 3 steps:

- (1) Write the *augmented matrix* of the system.
- (2) Use row operations to reduce (*working “top down”) the augmented matrix to echelon form (EF).
- (3) Use the (EF) to identify which variables are pivot variables (variables with pivots in their column), and which are free variables (variables without).
 - (3.5) Assign each free variable a “free parameter” letter. Though it doesn’t matter which letters you use, we usually use s , t , and u (if necessary).

Then we have a choice for the last “step”:

- (4A) From (EF), translate back to equations, and work *bottom up* “substituting” variables to get equations

$$(\text{pivot vars}) = (\#) + (\text{free variable}(s), \text{ if any}).$$

- (4B) From (EF), continue using row operations until the matrix is in reduced echelon form (REF). Then translate back to equations and “read off” equations

$$(\text{pivot vars}) = (\#) + (\text{free variable}(s), \text{ if any}).$$

- If we use (1)–(3) and then (4A), this method is called “*Gaussian elimination*”, or just “*the method of elimination*”.
- If we use (1)–(3) and then (4B), this method is “*Gauss-Jordan elimination*”.
- Both methods take roughly equal amounts of work/time, though I prefer reducing to (REF) and not worrying about substitution.

Facts/Tips/Tricks.

- (1) A system will be *inconsistent* (aka have no solutions) if the augmented matrix of the system has any (EF) (doesn’t have to be (REF)) that has a row that looks like

$$\left[\begin{array}{cc|c} 0 & 0 & (\neq 0) \end{array} \right], \quad \left[\begin{array}{ccc|c} 0 & 0 & 0 & (\neq 0) \end{array} \right], \quad \text{etc.,}$$

where “ $(\neq 0)$ ” just means any number that isn’t 0. This is the same as saying that the last column of the augmented matrix has a pivot. (Emphasis b/c if it’s not an augmented matrix of a system, i.e. it’s just a plain matrix, then we aren’t concerned with solutions).

- (2) A system has a *unique solution* if: The system is consistent, and in the augmented matrix of the system, every variable is a pivot variable. That is:

$$(\text{consistent}) + (\text{no free vars}) \implies (\text{unique solution}).$$

- (3) If a system has *infinitely many solutions* if: System is consistent (so it has at least one solution) and it has any free variables. That is:

$$(\text{consistent}) + (\text{at least one free var}) \implies (\text{infinitely many solutions}).$$

- (4) It's good to start with a "1" as your top pivot when reducing a matrix, so swap rows and/or scale them to simplify things if necessary. For example, starting with the matrix

$$\begin{bmatrix} 3 & 8 & 7 \\ 1 & 2 & 1 \\ 2 & 7 & 9 \end{bmatrix}$$

and trying to reduce/eliminate with

$$(R_2 \rightarrow R_2 - \tfrac{1}{3}R_1) : \quad \begin{bmatrix} 3 & 8 & 7 \\ 1 & 2 & 1 \\ 2 & 7 & 9 \end{bmatrix} \longrightarrow \begin{bmatrix} 3 & 8 & 7 \\ 0 & -\frac{2}{3} & -\frac{4}{3} \\ 2 & 7 & 9 \end{bmatrix}$$

is messier than first Swapping, then Adding:

$$\begin{aligned} (R_1 \leftrightarrow R_2) : \quad & \begin{bmatrix} 3 & 8 & 7 \\ 1 & 2 & 1 \\ 2 & 7 & 9 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 7 \\ 2 & 7 & 9 \end{bmatrix} \\ (R_2 \rightarrow R_2 - 3R_1) : \quad & \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 7 \\ 2 & 7 & 9 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 4 \\ 2 & 7 & 9 \end{bmatrix}. \end{aligned}$$

You will get the same answers/solutions either way (especially because you'll eventually end up at the one (REF)), but the path to get there can be messier/cleaner.